

Bounded Linear operators on Hilbert spaces (Chapter 8)

- Orthogonal projection operators.
- We revisit/extend the notion of Direct sum to X : vector space.
- Definition: $X \triangleq M \oplus N$ for M and N subspaces if $\forall x \in X$, there is $y \in M$ and $z \in N$ s.t. $x = y + z$ and decomposition unique.
- Uniqueness implies $M \cap N = \{0\}$. No orthogonality needed.
- For $X = M \oplus N$, we define $P: X \rightarrow X$ projection onto M along N by $Px = y$ in above decomposition.
- We verify that P is linear and $P^2 = P$ (idempotent)
- Definition A projection operator is a linear operator $X \rightarrow X$ such that $P^2 = P$.

- Theorem. X v.s.p. (a) If $P: X \rightarrow X$ projector, then $X = \text{ran } P \oplus \text{Ker } P$.
 (b) $X = M \oplus N$. Then $\exists P: X \rightarrow X$ projector with
 $M = \text{ran } P$ and $N = \text{Ker } P$

• proof (b) was done on previous page by defining $y = Px$ from $x = y + z$.

(a) Assume $y \in \text{ran } P$, i.e. $y = Px$. Then $P_y = P^2x = Px = y$.

So $y \in \text{ran } P \iff y = Py$.

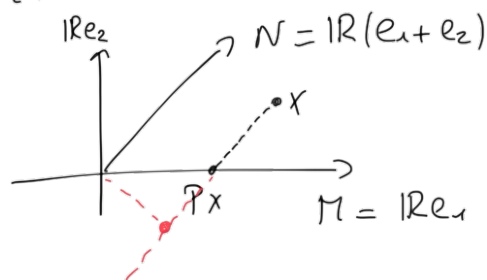
Now decompose $x = Px + (x - Px)$. $Px \in \text{ran } P$ and $x - Px \in \text{Ker } P$.

Is decomposition unique? If $x = y + z$ $y \in \text{ran } P$ and $z \in \text{Ker } P$,

then $Px = Py = y$ from above and $z = x - y = x - Px \square$

• N.B. $I - P$ projects onto N along M .

• N.B. no orthogonality here.



$$\mathbb{R}^2 = M \oplus N.$$

- On $\mathcal{H}$ Hilbert space, we have $\perp$ from $\langle \cdot, \cdot \rangle$.
- Projection theorem states $\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp$ when $\mathcal{M}$ closed subspace.
- Projection of $\mathcal{H}$ onto $\mathcal{M}$ along $\mathcal{M}^\perp$ is called orthogonal projection.
- Then: $\langle Px, y \rangle = \langle \underbrace{Px}_{\in \mathcal{M}}, \underbrace{y - Py + Py}_{\in \mathcal{M}^\perp} \rangle = \langle Px, Py \rangle = \langle x, Py \rangle$.

• Definition. An orthogonal projector on $\mathcal{H}$ is a linear operator $P: \mathcal{H} \rightarrow \mathcal{H}$ such that $P^2 = P$ and $\langle x, Py \rangle = \langle Px, y \rangle \forall x, y \in \mathcal{H}$.

• Proposition $\|P\| = 1$ unless $P = 0$ for orthogonal projectors.

• proof. If $\text{ran } P = \{0\}$ then $P = 0$.

• Otherwise, there is $x \in \mathcal{H}$ s.t. $Px \neq 0$.

• So for $y = Px$, $Py = y$ and $\|Py\| = \|y\| \Rightarrow \|P\| \geq 1$
(no need for orthogonality here).

• now, for $Px \neq 0$, $\|Px\| = \frac{\langle Px, Px \rangle}{\|Px\|} \stackrel{c.s.}{=} \frac{\langle x, P^2x \rangle}{\|Px\|} \leq \frac{\|x\| \|Px\|}{\|Px\|} \leq \|x\|$. $\square$

Theorem. $\mathcal{H}$ Hilbert. (a) P $\perp$ projection. Then $\text{ran } P$ is closed and $\mathcal{H} = \text{ran } P \oplus \text{Ker } P$.
 (b) $\mathcal{H} = \mathcal{M} \oplus \mathcal{M}^\perp$. Then $\exists ! P$ orthogonal projection s.t. $\mathcal{M} = \text{ran } P$.

Proof (b) was done in construction of projector. P unique since $Px = x$ on $\mathcal{M}$ and $Px = 0$ on $\mathcal{M}^\perp$.

(a) We already know from previous theorem that $\mathcal{H} = \text{ran } P \oplus \text{Ker } P$.

Since P is orthogonal, $\text{ran } P \perp \text{Ker } P$: $x = Px + x - Px$; $\langle Px, x - Px \rangle = 0$.

Therefore: $\text{ran } P = (\text{Ker } P)^\perp$ since decomposition unique. So $\text{ran } P$ closed.

Rem. { If P is continuous (bounded), then $\text{ran } P \ni y_n \rightarrow y \Rightarrow y \leftarrow y_n = Py_n \rightarrow Py = y$.
 we know that $\perp$ proj. are bounded and this gives another proof that $\text{ran } P$ is closed.
 There are unbounded projectors whose range is not closed.

Rem For P as above, then $I - P$ is $\perp$ projector onto $\mathcal{M}^\perp$.

Examples $(\mathbb{P} f)(x) = \frac{f(x) + f(-x)}{2}$ $\perp$ proj. in $L^2(\mathbb{R})$ onto even functions

. A measurable set in $\mathbb{R}$ (or general $(X, \mathcal{A}, \mu)$) $\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$

$(\mathbb{P}_A f)(x) = \chi_A(x) f(x)$ orthogonal projection of $L^2(\mathbb{R})$ onto functions with support in $\bar{A}$.

. Rank-one projection. $\mathbb{P}_u(x) = \langle u, x \rangle u$ $\mathbb{P}_u = u \otimes u = |u\rangle \langle u|$
for $\|u\| = 1$.

. $\mathcal{H} = L^2(\mathbb{T})$ $\mathbb{P} f = \langle f \rangle = \frac{1}{2\pi} \int_{\mathbb{T}} f(x) dx$ projection onto constants

$f(x) = \langle f \rangle + \tilde{f}(x)$ $\langle \langle f \rangle, \tilde{f} \rangle = 0$.
fluctuations

• Dual of a Hilbert space

Recall linear form $\varphi: \mathcal{H} \rightarrow \begin{cases} \mathbb{R} \\ \mathbb{C} \end{cases}$ $|\varphi(x)| \leq M \|x\|$.

$$\|\varphi\|_{\mathcal{H}^*} = \sup_{\substack{x \in \mathcal{H} \\ \|x\|=1}} |\varphi(x)|.$$

For $y \in \mathcal{H}$, we define $\varphi_y(x) := \langle y, x \rangle \quad \forall x \in \mathcal{H}$.

$|\varphi_y(x)| \leq \|y\| \|x\|$ and $|\varphi_y(y)| = \|y\|^2$ so $\|\varphi_y\|_{\mathcal{H}^*} = \|y\|_{\mathcal{H}}$ and $\varphi_y \in \mathcal{H}^*$.

Theorem [Riesz representation]. Let $\varphi \in \mathcal{H}^*$ bounded linear form. Then

there is a unique $y \in \mathcal{H}$ s.t. $\varphi(x) = \langle y, x \rangle \quad \forall x \in \mathcal{H}$.

$J: \mathcal{H} \rightarrow \mathcal{H}^*$ defined as $Jy = \varphi_y$ is isometry between $\mathcal{H}$ and $\mathcal{H}^*$; $\|y\|_{\mathcal{H}} = \|\varphi_y\|_{\mathcal{H}^*}$.

$\therefore$ therefore $\mathcal{H} \cong \mathcal{H}^*$

• Note $(L^p)^* = L^q$ $\frac{1}{p} + \frac{1}{q} = 1$ with $L^2 = (L^2)^*$ $1 < p < \infty$.

but $L^p \not\cong L^q$ for $p \neq q$ (not easy to prove).

Proof. If $\varphi(x) \equiv 0$, take $y=0$. If not, there is $z \in (\ker \varphi)^\perp$ s.t. $z \neq 0$
 since $\mathcal{H} = \ker \varphi \oplus (\ker \varphi)^\perp$ by projection theorem.

Let $P: \mathcal{H} \rightarrow \mathcal{H}$ be defined as $Px = \frac{\varphi(x)}{\varphi(z)} z$ ($\varphi(z) \neq 0$).

P is linear and $P^2 x = \frac{\varphi(x)}{\varphi(z)} \overbrace{Pz}^{=z} = Px$. So, $\mathcal{H} = \text{ran } P \oplus \ker P$.

$Px=0 \Rightarrow \varphi(x)=0$ so $\ker P = \ker \varphi$. $\text{Ran } P = \mathbb{C}z = \{\alpha z, \alpha \in \mathbb{C}\}$.

Since $z \in (\ker \varphi)^\perp = (\ker P)^\perp$, then $\text{Ran } P \perp \ker P$ and P is orthogonal.

Thus, $\mathcal{H} = \text{ran } P \oplus \ker P = \mathbb{C}z \oplus \ker \varphi$ orthogonal direct sum.

Let $x \in \mathcal{H}$. Then $x = \alpha z + n$ with $\langle z, n \rangle = 0 \Rightarrow \alpha = \frac{\langle z, x \rangle}{\|z\|^2}$.

$\varphi(x) = \alpha \varphi(z) + \varphi(n) = \alpha \varphi(z) = \varphi(z) \frac{\langle z, x \rangle}{\|z\|^2} = \left\langle \frac{\varphi(z)z}{\|z\|^2}, x \right\rangle$.

Choose $y := \frac{\varphi(z)z}{\|z\|^2}$ so that $\varphi(x) = \langle y, x \rangle \forall x \in \mathcal{H}$.

Uniqueness: $\varphi_{y_1}(x) = \varphi_{y_2}(x) = \langle y_1, x \rangle = \langle y_2, x \rangle \quad \forall x$

Take $x = y_1 - y_2 \Rightarrow y_1 = y_2$. □

• Adjoint operator. Let A bounded on $\mathcal{H}$.

We want to construct A^* such that $\langle x, Ay \rangle = \langle A^*x, y \rangle \quad \forall x, y \in \mathcal{H}$.

• Fix $x \in \mathcal{H}$. $\langle x, Ay \rangle = \varphi_x(y)$ a bounded linear form $\|\varphi_x\| \leq \|A\| \|x\|$.

by Riesz, $\varphi_x(y) = \langle z, y \rangle$, unique $z \in \mathcal{H}$. Define $A^*x = z$.

• verify A^* linear. | Assume $\langle A_1^*x, y \rangle = \langle A_2^*x, y \rangle \Leftrightarrow \langle (A_1^* - A_2^*)x, y \rangle = 0 \quad \forall y$.

Uniqueness: | This implies $(A_1^* - A_2^*)x = 0 \quad \forall x \Leftrightarrow A_1^* = A_2^*$.

• So A^* constructed and bounded (check).

• Properties: $A^{**} = A \quad (AB)^* = B^*A^* \quad (\text{check!})$

Example. $\mathbb{R}^n$: $A^* = A^T \quad (A^T)_{ij} = A_{ji}$. $\mathbb{C}^n$: $A^* = \overline{A^T} \quad A^*_{ij} = \overline{A_{ji}}$.

$Kf(x) = \int_0^1 k(x, y) f(y) dy$ on $L^2(0, 1)$. Then: $K^*f(x) = \int_0^1 \overline{k(y, x)} f(y) dy$.

(check!)

• Role of adjoint operator in solving equations.

• $Ax = y$ $A: \mathcal{H} \rightarrow \mathcal{H}$ bounded. Assume $A^*z = 0$ for $z \neq 0$

$$\text{Then } \langle y, z \rangle = \langle Ax, z \rangle = \langle x, A^*z \rangle = 0$$

Thus, $\langle y, z \rangle = 0$ is necessary condition for $Ax=y$ to have a solution.

Theorem $A: \mathcal{H} \rightarrow \mathcal{H}$ bounded. Then: $\overline{\text{ran } A} = (\text{Ker } A^*)^\perp$
 $\text{ker } A = (\text{ran } A^*)^\perp$

$$\mathcal{H} = \overline{\text{ran } A} \oplus \text{ker } A^* = \overline{\text{ran } A^*} \oplus \text{ker } A.$$

• Proof. Let $x \in \text{ran } A$; $x = Ay$ and $z \in \text{Ker } A^*$. Then:

$$\langle x, z \rangle = \langle Ay, z \rangle = \langle y, A^*z \rangle = 0. \quad \text{So } \text{ran } A \subset (\text{ker } A^*)^\perp \\ \Rightarrow \overline{\text{ran } A} \subset (\text{ker } A^*)^\perp.$$

$$\text{Let } x \in (\text{ran } A)^\perp. \quad 0 = \langle Ay, x \rangle = \langle y, A^*x \rangle \quad \forall y \Rightarrow A^*x = 0$$

$$\text{So } (\text{ran } A)^\perp \subset \text{ker } A^* \Rightarrow (\text{ker } A^*)^\perp \subset \text{ran } A^{\perp\perp} = \overline{\text{ran } A}.$$

• This is $\overline{\text{ran } A} = (\text{Ker } A^*)^\perp$. Thus $\text{ker } A^* = (\text{ran } A)^\perp$
and $A^{**} = A$ concludes $\square$

• Theorem. A bdd with closed range. Then $Ax=y$ has a solution iff $y \in (\text{Ker } A^*)^\perp$.

proof: $\mathcal{U} = \overline{\text{ran } A} \oplus \text{Ker } A^* = \text{ran } A \oplus \text{Ker } A^*$ orthogonal direct sum.

• recall $\|Ax\| \geq c\|x\|$ for $c > 0 \Rightarrow \text{ran } A$ closed (and $\text{ker } A = \{0\}$).

• Definition Fredholm alternative. $A: \mathcal{U} \rightarrow \mathcal{U}$ satisfies F.A. if

(a) either $Ax=0$ and $A^*x=0$ have only 0 as solutions; and then $Ax=y$ and $A^*x=y$ have unique solutions

(b) or $Ax=0$ and $A^*x=0$ admit finite dimensional spaces of solutions of the same positive (≥ 1) dimension; and then $Ax=y$ has solutions iff $y \perp z \forall A^*z=0$ and $A^*x=y$ _____ $Az=0$.

• This is a definition. operators $\mathbb{C}^n \rightarrow \mathbb{C}^n$ satisfy F.A.

Since $\text{ran } A = \overline{\text{ran } A}$ and $\dim \text{ran } A = \dim \text{ran } A^*$.

• This does not always hold in ∞ -dim. Two possible issues: (i) $\text{ran } A$ not closed and (ii) $\dim \text{ker } A \neq \dim \text{Ker } A^*$.

• Examples . $L^2(0,1) \rightarrow L^2(0,1)$

$$(Mf)(x) = x f(x) . \quad M^* = M \quad \text{ran } M \neq \overline{\text{ran } M} = L^2(0,1) .$$

• $S: \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N}) \quad S(x_1, \dots, x_n, \dots) = (0, x_1, x_2, \dots)$ Shift.

$$\text{ran } S = \{x \text{ s.t. } x_1 = 0\} \text{ closed.} \quad \text{Ker } S = \{0\}.$$

$$T = S^* \quad T(x_1, \dots) = (x_2, \dots)$$

$$\text{Ker } T = \{1, 0, \dots, 0, \dots\} \text{ one-dimensional.}$$

$$\dim \text{Ker } S^* = 1 \neq \dim \text{Ker } S = 0. \quad \text{F.A. not satisfied.}$$

• Def A bounded is Fredholm if a) $\text{ran } A$ closed; b) $\text{Ker } A$ and $\text{Ker } A^*$ finite dimensional.

• Index of F.O. $\text{Index } A = \dim \text{Ker } A - \dim \text{Ker } A^*$

• F.A. satisfied for F.O. iff $\text{Index } A = 0$

• Note A Fredholm, K compact. Then $A+K$ Fredholm and
 $\text{Index } (A+K) = \text{Index } (A).$

Self-adjoint and unitary operators

Definition: $A: \mathcal{H} \rightarrow \mathcal{H}$ is self-adjoint if $A^* = A$.

Define the sesquilinear form $a: \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ by $a(x, y) = \langle x, Ay \rangle$.

If $A = A^*$, then $a(x, y) = \overline{a(y, x)}$ is Hermitian symmetric

Define quadratic form $q(x) = a(x, x) = \langle x, Ax \rangle \in \mathbb{R}$ when $A = A^*$.

Definition: A is non-negative if A self-adjoint and $\langle x, Ax \rangle \geq 0$.

A is positive-definite if $\langle x, Ax \rangle > 0$ when $x \neq 0$.

If A is positive definite (and bounded), then

$(x, y)_A := \langle x, Ay \rangle$ is an inner product.

Lemma. A bounded self-adjoint. Then $\|A\| = \sup_{\|x\|=1} |\langle x, Ax \rangle|$.

Proof. $\langle \cdot, \cdot \rangle \rightarrow \mathbb{R}$ here (see [H.N.] for $\rightarrow \mathbb{C}$).

↙ C.S. with $y \|Ax\| = Ax$
or Hahn-Banach (char. norm)

$$\alpha = \sup_{\|x\|=1} \langle x, Ax \rangle. \quad \|A\| = \sup_{\|x\|=1} \|Ax\| = \sup_{\|x\|=\|y\|=1} |\langle y, Ax \rangle| \geq \alpha$$

$$A = A^* \Rightarrow 4 \langle y, Ax \rangle = \langle x+y, A(x+y) \rangle - \langle x-y, A(x-y) \rangle$$

$$\text{so, } |4 \langle y, Ax \rangle| \leq \alpha (\|x+y\|^2 + \|x-y\|^2) = 2\alpha (\|x\|^2 + \|y\|^2)$$

$$\Rightarrow |\langle y, Ax \rangle| \leq \alpha \text{ since } \|x\| = \|y\| = 1 \Rightarrow \|A\| \leq \alpha \quad \square$$

Corollary A bounded. Then $\|A^*A\| = \|A\|^2$. When $A = A^*$, then $\|A\|^2 = \|A^2\|$.

$$\text{proof: } (A^*A)^* = A^*A \Rightarrow \|A^*A\| = \sup_{\|x\|=1} |\langle x, A^*Ax \rangle| = \sup_{\|x\|=1} \|Ax\|^2 = \|A\|^2 \quad \square$$

Definition. Unitary operator. $U: \mathcal{H}_1 \rightarrow \mathcal{H}_2$ is unitary if it is invertible and $\langle Ux, Uy \rangle_{\mathcal{H}_2} = \langle x, y \rangle_{\mathcal{H}_1}$.

. Then $\mathcal{H}_1$ and $\mathcal{H}_2$ are isomorphic: invertible map preserving inner product.

. $U: \mathcal{H} \rightarrow \mathcal{H}$ is unitary iff $U^* U = U U^* = \text{Id on } \mathcal{H}$, i.e., $U^{-1} = U^*$.

. Definition S is skew self-adjoint if $S^* = -S$.

Then $S = iA$ with A self-adjoint.

. Definition $T: \mathcal{H} \rightarrow \mathcal{H}$ is normal if $T^* T = T T^*$.

. Check that $T: \mathcal{H} \rightarrow \mathcal{H}$ s.A., skew s.A. or unitary

$\Rightarrow T$ normal.

. This is the largest class of operators that can be "diagonalized".

. Weak convergence in Hilbert spaces.

. $\mathcal{H}$ Hilbert. $\mathcal{H}^* \cong \mathcal{H}$ so weak and weak* convergences are "the same".

. $x_n \rightarrow x \Leftrightarrow \langle x_n, y \rangle \rightarrow \langle x, y \rangle \quad \forall y \in \mathcal{H}.$

Theorem Uniform boundedness (Banach Steinhaus). $\varphi_n: X \rightarrow \mathbb{C}; X$ Banach.

$\{\varphi_n\}_{n \in \mathbb{N}}$ linear forms such that $|\varphi_n(x)| \leq C(x)$ for all $x \in \mathcal{H}$.
(uniform in n)

Then there is a constant C' s.t. $\|\varphi_n\| \leq C'$ (uniform in n).

Proof see [H.W.]. (i) show φ_n uniformly bounded on any ball $\Rightarrow$ uniformly bounded.

(ii) If not unif. bdd, construct $|\varphi_{n_k}(x_k)| > k$ for x_k Cauchy $\rightarrow \bar{x}$.

Obtain contradiction on $|\varphi_n(\bar{x})| \leq C(\bar{x})$. $\square$

. That φ_n map into finite dimensional spaces simplifies the proof.

Theorem: $x_n \in \mathcal{H}$. $\mathcal{D}$ dense in $\mathcal{H}$.

$$\text{Then } x_n \rightharpoonup x \quad \text{iif} \quad \begin{cases} (a) \quad \|x_n\| \leq M \\ (b) \quad \langle x_n, y \rangle \rightarrow \langle x, y \rangle \quad \forall y \in \mathcal{D}. \end{cases}$$

• note that the above implies that $x_n \rightharpoonup x \Rightarrow \|x_n\|$ uniformly bounded.

• proof $\Rightarrow x_n \rightharpoonup x$, $\varphi_n(x) = \langle x_n, x \rangle$ converges to $\|x\|^2 \Rightarrow |\varphi_n(x)| \leq C(x)$.
By Banach-Steinhaus, $\|x_n\| = \|\varphi_n\| \leq C$. Done

$\Leftarrow z \in \mathcal{H}$ with $\|y - z\| \leq \varepsilon$ for $y = y(\varepsilon) \in \mathcal{D}$.

$$\begin{aligned} \text{Then } |\langle x_n - x, z \rangle| &= |\langle x_n - x, z - y + y \rangle| \leq (\|x_n\| + \|x\|) \varepsilon + |\langle x_n - x, y \rangle| \\ &\leq (\|x\| + M) \varepsilon + \varepsilon = C\varepsilon \text{ when } n \text{ large.} \end{aligned}$$

so $\langle x_n - x, z \rangle \rightarrow 0 \quad \forall z$ and $x_n \rightarrow x$.